

# HireSense AI

## AI-Powered Modular Placement & Recruitment Automation Platform

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Detailed Project & Module Specification Report  
30 System Modules · Features · Architecture Notes · Suggested Build Roadmap

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# 1. Project Overview

HireSense AI is a comprehensive, AI-powered recruitment automation platform designed to digitalize and automate the complete hiring lifecycle end to end. It provides a centralized ecosystem where recruiters can post job opportunities, configure fully customized recruitment workflows, evaluate candidates through multiple assessment rounds, and generate intelligent, data-backed hiring decisions. On the other side of the marketplace, students can build professional profiles, apply for jobs, participate in online assessments, attend AI-assisted interviews, and receive personalized feedback and reports.

The platform is built around four guiding principles: **automation** (minimizing manual recruiter effort at every stage), **fairness** (standardized, bias-resistant evaluation criteria), **transparency** (clear status tracking and feedback for candidates), and **scalability** (an architecture capable of handling large-scale placement drives with concurrent assessments). Crucially, the system does not impose a single fixed hiring process — recruiters configure their own pipeline per job, and the Workflow Engine dynamically executes whichever combination of rounds they choose.

The system is organized into 30 modules, grouped below by functional area. Each module is documented in detail in Section 3, including every listed feature and its purpose within the platform.

## Functional Groupings

|                                |                                                                                                    |
|--------------------------------|----------------------------------------------------------------------------------------------------|
| Platform Foundation            | Modules 1, 2, 26, 27, 28, 30 — authentication, public site, security, storage, logging, deployment |
| User & Organization Management | Modules 3, 4, 5 — student profiles, recruiter/company profiles, admin governance                   |
| Job & Application Lifecycle    | Modules 6, 7, 8, 9, 10, 25 — job postings, workflow configuration, resumes, applications           |
| Assessment Rounds              | Modules 11, 12, 13, 14 — aptitude, coding, sandboxing, group discussion                            |
| AI Interview System            | Modules 15, 16, 17, 18, 19 — interviews, question generation, NLP follow-ups, integrity monitoring |
| Decisioning & Reporting        | Modules 20, 21, 22, 23, 24 — evaluation, shortlisting, reports, notifications, analytics           |
| Future Roadmap                 | Modules 9 (future), 29 — advanced AI microservices                                                 |

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|           |                                   |
|-----------|-----------------------------------|
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*Also included: Section 3 (Detailed Modules), Section 4 (Suggested Technology Stack), Section 5 (Suggested Build Roadmap), Section 6 (Closing Notes).*

### 3. Detailed Module Specifications

#### Module 1 — Authentication & Authorization

*Secure identity management for Students, Recruiters, and Admins*

This module forms the security backbone of HireSense AI, governing how every user type gains access to the platform. It ensures that only verified individuals can create accounts and interact with sensitive recruitment data, while role-based rules keep students, recruiters, and administrators confined to the actions appropriate to their role. The module combines standard credential-based login with modern token-based session handling so that the platform remains both secure and scalable as the user base grows.

**Features (11)**

|                                         |                                                                                                                                              |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Student Registration</b>             | Allows students to create an account using email, password, and basic profile details, forming the entry point into their placement journey. |
| <b>Recruiter Registration</b>           | Enables company representatives to sign up as recruiters, after which their organization can be verified before posting jobs.                |
| <b>Admin Login</b>                      | A restricted, separately secured login path for platform administrators to access system-wide controls.                                      |
| <b>Email OTP Verification</b>           | Sends a one-time password to the registered email to confirm ownership of the account before activation.                                     |
| <b>JWT Authentication</b>               | Uses signed JSON Web Tokens to authenticate API requests statelessly and securely across sessions.                                           |
| <b>Refresh Tokens</b>                   | Issues long-lived refresh tokens that allow short-lived access tokens to be renewed without forcing the user to log in again.                |
| <b>Forgot Password</b>                  | Lets users request a secure password-reset link or code when they are unable to log in.                                                      |
| <b>Reset Password</b>                   | Provides a secure flow to set a new password after identity verification during a forgot-password request.                                   |
| <b>Change Password</b>                  | Allows logged-in users to update their password from their account settings.                                                                 |
| <b>Session Management</b>               | Tracks active sessions per user, supporting logout, session expiry, and multi-device handling.                                               |
| <b>Role-Based Access Control (RBAC)</b> | Restricts each API endpoint and UI view based on the user's role (Student, Recruiter, Admin), preventing unauthorized access.                |

## Module 2 — Landing Website

*The public face of the platform that converts visitors into registered users*

The Landing Website module is the first touchpoint for anyone discovering HireSense AI. It communicates the platform's value proposition to both students and recruiters, builds trust through statistics and testimonials, and funnels visitors toward registration or login. Since this is the marketing and information layer, it is built to be fast, responsive, and SEO-friendly, independent of the authenticated application areas.

### Features (10)

|                    |                                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------|
| Hero Section       | A prominent introductory banner with a headline, sub-text, and primary call-to-action buttons for sign-up.              |
| About Platform     | A section explaining what HireSense AI does, its mission, and the problem it solves for placement drives.               |
| Features           | A showcase grid or list summarizing the platform's key capabilities such as AI interviews and automated evaluation.     |
| Statistics         | Displays platform metrics such as number of students placed, companies onboarded, or jobs posted to build credibility.  |
| Recruiter Benefits | Highlights the value proposition specifically for recruiters, such as time saved and quality of shortlisted candidates. |
| Student Benefits   | Highlights the value proposition specifically for students, such as personalized feedback and fair evaluation.          |
| Testimonials       | Displays quotes or reviews from students and recruiters who have used the platform successfully.                        |
| FAQs               | An accordion-style list answering common questions about registration, pricing, and platform usage.                     |
| Contact Form       | Allows visitors to send inquiries to the platform team directly from the website.                                       |
| Login & Register   | Prominent navigation entry points that route users into the authentication flows of Module 1.                           |

## Module 3 — Student Management

*Enables students to build a complete, verifiable professional profile*

This module gives students a structured space to consolidate everything recruiters need to evaluate them, replacing scattered resumes and disconnected profiles with a single living dashboard. A well-completed profile directly powers downstream modules such as Resume Ranking AI and the Evaluation Engine, so the platform encourages high profile completeness through guided prompts and completion tracking.

### Features (12)

|                             |                                                                                                               |
|-----------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Personal Information</b> | Captures core identity details such as name, contact information, location, and profile photo.                |
| <b>Academic Details</b>     | Records degree, institution, branch, year of graduation, and academic performance (CGPA/percentage).          |
| <b>Skills</b>               | A tagged list of technical and soft skills that feeds directly into resume-job matching algorithms.           |
| <b>Certifications</b>       | Lets students add professional certifications with issuing authority and validity, strengthening credibility. |
| <b>Projects</b>             | A structured section to showcase academic or personal projects with descriptions, tech stack, and links.      |
| <b>Work Experience</b>      | Captures internships and prior employment with role, duration, and responsibilities.                          |
| <b>GitHub</b>               | Links the student's GitHub profile so recruiters can review code quality and contribution history.            |
| <b>LinkedIn</b>             | Links the student's LinkedIn profile for professional networking verification.                                |
| <b>Portfolio</b>            | Allows linking of an external portfolio website showcasing additional work.                                   |
| <b>Resume Upload</b>        | Lets students upload their resume file, which is stored and later parsed by the Resume Management module.     |
| <b>Profile Completion</b>   | A progress indicator that nudges students to fill in all sections for a stronger application.                 |
| <b>Dashboard</b>            | A personalized home screen summarizing applications, upcoming assessments, interviews, and notifications.     |

## Module 4 — Recruiter Management

*Gives recruiters full control over their organization's hiring presence*

Recruiter Management equips company representatives with the tools to represent their organization on the platform, configure how their hiring process runs, and track candidates through every stage. Because trust is critical in a hiring marketplace, company identity is verified before recruiters gain full posting privileges.

### Features (7)

|                          |                                                                                                         |
|--------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Company Profile</b>   | Stores organization details such as name, industry, size, logo, and description for candidates to view. |
| <b>Recruiter Profile</b> | Captures the individual recruiter's details, designation, and contact information within the company.   |

|                           |                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------|
| Company Verification      | An admin-reviewed process confirming the legitimacy of the organization before it can post jobs.   |
| Dashboard                 | A recruiter's control center summarizing active jobs, applicants, and pipeline status at a glance. |
| Job Management            | Central hub to create, edit, publish, and close job postings (detailed further in Module 6).       |
| Recruitment Configuration | Access point to define the hiring workflow and rounds for each job (detailed in Module 7).         |
| Candidate Tracking        | A pipeline view showing where each applicant stands across the configured recruitment stages.      |

## Module 5 — Admin Panel

*Centralized oversight and governance for the entire platform*

The Admin Panel gives platform administrators the visibility and controls needed to keep HireSense AI trustworthy, compliant, and healthy at scale. From approving new companies to auditing suspicious activity, this module acts as the control tower overseeing students, recruiters, jobs, and system behavior.

### Features (8)

|                        |                                                                                               |
|------------------------|-----------------------------------------------------------------------------------------------|
| User Management        | Allows admins to view, search, suspend, or delete student and recruiter accounts.             |
| Recruiter Verification | Workflow for admins to review and approve or reject recruiter sign-up requests.               |
| Company Approval       | Formal approval step confirming a company's legitimacy before it can operate on the platform. |
| Job Monitoring         | Lets admins review all job postings for policy compliance and quality.                        |
| Platform Analytics     | High-level dashboards showing platform-wide usage, growth, and engagement metrics.            |
| Reports                | Generates administrative reports on hiring activity, user growth, and system health.          |
| Activity Logs          | A searchable trail of significant actions taken across the platform for accountability.       |
| System Configuration   | Global settings such as feature toggles, evaluation defaults, and platform parameters.        |



## Module 6 — Job Portal

*The core marketplace connecting job postings with student applicants*

The Job Portal is where recruitment activity becomes visible to students. Recruiters publish structured job listings, and students discover, filter, and apply to opportunities that match their profile. This module ties directly into the Recruitment Workflow Configuration (Module 7), since each job's application journey depends on the process the recruiter has configured for it.

### Features (10)

|                             |                                                                                                        |
|-----------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Create Job</b>           | Lets recruiters draft a new job posting with role, description, eligibility, and compensation details. |
| <b>Edit Job</b>             | Allows updates to an existing job listing before or during the application window.                     |
| <b>Delete Job</b>           | Permanently removes a job posting, typically restricted to jobs with no active applications.           |
| <b>Publish Job</b>          | Makes a drafted job visible to students on the portal.                                                 |
| <b>Close Job</b>            | Stops new applications once the recruiter has enough candidates or the deadline has passed.            |
| <b>Draft Jobs</b>           | Saves incomplete job postings privately before they are ready to publish.                              |
| <b>Search Jobs</b>          | Lets students filter and search postings by role, skill, location, or company.                         |
| <b>Bookmark Jobs</b>        | Allows students to save jobs of interest to revisit later.                                             |
| <b>Apply Jobs</b>           | The core action letting a student submit their profile and resume against a specific job.              |
| <b>Application Tracking</b> | Gives students real-time visibility into the status of each application they have submitted.           |

## Module 7 — Recruitment Workflow Configuration

*Lets each recruiter design a bespoke hiring pipeline*

This module is central to HireSense AI's flexibility: rather than forcing every company through an identical process, recruiters can turn assessment stages on or off, decide how each stage is weighted, and set cutoffs — all before recruitment begins. This configuration is later read by the Workflow Engine (Module 25) to dynamically build the actual candidate pipeline for that job.

### Features (8)

|                                        |                                                                                                        |
|----------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Resume Screening</b>                | Toggle to enable an initial automated or manual screening of resumes before further rounds.            |
| <b>Enable/Disable Aptitude</b>         | Lets the recruiter decide whether an aptitude test round is part of this job's pipeline.               |
| <b>Enable/Disable Coding</b>           | Lets the recruiter decide whether a coding assessment round is included.                               |
| <b>Enable/Disable Group Discussion</b> | Lets the recruiter decide whether a group discussion round is included.                                |
| <b>AI Interview or Human Interview</b> | Choice between an AI-conducted interview or a recruiter-scheduled human interview for the final round. |
| <b>Cutoff Configuration</b>            | Set minimum scores required at each stage for a candidate to progress to the next round.               |
| <b>Weightage Configuration</b>         | Assign relative importance (weights) to each stage's score when computing the final evaluation.        |
| <b>Hiring Templates</b>                | Save a configured workflow as a reusable template for future job postings.                             |

## Module 8 — Resume Management

*Handles the full lifecycle of a candidate's resume document*

This module manages resumes as first-class documents within the platform — from upload to parsing to long-term storage. Reliable resume handling is a prerequisite for both the Resume Ranking AI (Module 9) and the AI Question Generator (Module 16), both of which depend on accurately extracted resume content.

### Features (7)

|                           |                                                                                                       |
|---------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Resume Upload</b>      | Accepts resume files (typically PDF/DOCX) from students with format and size validation.              |
| <b>Resume Preview</b>     | Renders the uploaded resume in-browser so students and recruiters can review it without downloading.  |
| <b>Resume Storage</b>     | Securely stores resume files in a managed file storage system with access control.                    |
| <b>Resume Replacement</b> | Allows students to upload a newer version of their resume, replacing the previous file.               |
| <b>Resume Download</b>    | Lets recruiters and students download the resume file when needed.                                    |
| <b>Resume Parsing</b>     | Extracts structured data (skills, education, experience) from unstructured resume text automatically. |
| <b>Resume History</b>     | Keeps a version history of resumes a student has uploaded over time.                                  |

## Module 9 — Resume Ranking AI (Future)

*AI-driven scoring of resumes against job requirements*

Marked as a future enhancement, this module envisions an intelligent layer that automatically evaluates how well a candidate's resume aligns with a specific job description. By quantifying fit across multiple dimensions, it helps recruiters prioritize review time on the most relevant candidates and can feed directly into the Stage-wise Shortlisting module.

### Features (6)

|                                 |                                                                                             |
|---------------------------------|---------------------------------------------------------------------------------------------|
| <b>Skill Matching</b>           | Compares candidate skills against required job skills to produce a match percentage.        |
| <b>Education Matching</b>       | Evaluates whether the candidate's academic background meets the job's educational criteria. |
| <b>Experience Matching</b>      | Assesses relevant work or project experience against the experience expected for the role.  |
| <b>Job Description Matching</b> | Uses NLP/semantic comparison between resume content and the full job description text.      |
| <b>Resume Score</b>             | Aggregates all matching dimensions into a single composite score for the resume.            |
| <b>Resume Ranking</b>           | Ranks all applicants for a job in order of their resume score to streamline shortlisting.   |

## Module 10 — Job Application Management

*Tracks every candidate's journey from application to outcome*

This module is the system of record for applications, maintaining an accurate, real-time picture of where each candidate stands. It bridges the Job Portal (where a student applies) and the Evaluation Engine (which determines the outcome), ensuring nothing falls through the cracks across potentially thousands of concurrent applications.

### Features (6)

|                        |                                                                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Apply</b>           | Records a student's submission for a specific job, linking their profile and resume snapshot.                |
| <b>Withdraw</b>        | Allows a student to voluntarily withdraw an active application.                                              |
| <b>Status Tracking</b> | Maintains and displays the current stage of an application (e.g., Applied, Shortlisted, Rejected, Selected). |

|                             |                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------|
| <b>Resume Submission</b>    | Attaches the specific resume version used for that application.                                 |
| <b>Recruiter View</b>       | Gives recruiters a consolidated list of all applicants for a job with quick access to profiles. |
| <b>Application Timeline</b> | Displays a chronological history of status changes for a given application.                     |

## Module 11 — Aptitude Test Module

*Automated, timed assessment of logical and quantitative ability*

The Aptitude Test Module delivers standardized, objective testing that is typically the first filter in a technical hiring pipeline. It is designed for scale, using randomized question sets and automated evaluation so hundreds of candidates can be assessed fairly and simultaneously, with results feeding straight into Stage-wise Shortlisting.

### Features (7)

|                          |                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>MCQs</b>              | Presents multiple-choice questions covering quantitative aptitude, logical reasoning, and verbal ability. |
| <b>Timer</b>             | Enforces a time limit for the overall test or per-question, displayed live to the candidate.              |
| <b>Random Questions</b>  | Serves a randomized subset of questions from a question bank to reduce cheating and repetition.           |
| <b>Difficulty Levels</b> | Tags questions by difficulty so recruiters can tune the overall test difficulty for a role.               |
| <b>Auto Evaluation</b>   | Automatically scores objective answers immediately upon submission.                                       |
| <b>Cutoff</b>            | Applies the recruiter-configured minimum score needed to pass this round.                                 |
| <b>Result Generation</b> | Produces a detailed score breakdown for each candidate upon test completion.                              |

## Module 12 — Coding Assessment Module

*Evaluates real programming ability in a controlled online environment*

This module lets candidates write and run actual code against test cases, giving recruiters an objective measure of programming proficiency rather than relying on self-reported skills. It works closely with the Coding Sandbox (Module 13) to execute untrusted candidate code safely at scale.

### Features (8)

|                               |                                                                                                |
|-------------------------------|------------------------------------------------------------------------------------------------|
| <b>Online IDE</b>             | A browser-based code editor with syntax highlighting where candidates write their solutions.   |
| <b>Multi-language Support</b> | Allows candidates to choose from multiple programming languages for a given problem.           |
| <b>Test Cases</b>             | Visible sample test cases that help candidates validate their solution logic.                  |
| <b>Hidden Test Cases</b>      | Additional test cases not shown to candidates, used to verify solution correctness rigorously. |
| <b>Docker Sandbox</b>         | Executes submitted code inside isolated Docker containers for security (see Module 13).        |
| <b>Time Limit</b>             | Restricts how long a submitted program is allowed to run before being terminated.              |
| <b>Memory Limit</b>           | Restricts the memory a submitted program can consume during execution.                         |
| <b>Automatic Evaluation</b>   | Compares program output against expected output for each test case to compute a score.         |

## Module 13 — Coding Sandbox

*A hardened execution environment for running untrusted candidate code*

Because the platform executes arbitrary code submitted by candidates, the Coding Sandbox exists purely to make that execution safe. It isolates every run from the host system and from other candidates' submissions, protecting platform infrastructure from malicious or accidental resource abuse.

### Features (6)

|                            |                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------|
| <b>Docker Isolation</b>    | Runs each code submission inside its own disposable, isolated container.                      |
| <b>Network Restriction</b> | Blocks outbound and inbound network access from within the execution container.               |
| <b>CPU Limits</b>          | Caps the CPU resources a single submission can consume to prevent abuse.                      |
| <b>Memory Limits</b>       | Caps the RAM a container can use, terminating runs that exceed the limit.                     |
| <b>File Restrictions</b>   | Restricts filesystem access so submitted code cannot read or write outside its sandbox.       |
| <b>Safe Code Execution</b> | The overall guarantee that no submission can compromise the host system or other users' data. |

## Module 14 — Group Discussion Module

*An optional collaborative round to assess communication and teamwork*

Group Discussion (GD) is an optional recruitment round that recruiters can enable through the Workflow Configuration module. It evaluates soft skills — articulation, listening, leadership, and collaboration — that are difficult to measure through tests alone, and requires scheduling and coordination among multiple candidates.

### Features (5)

|                              |                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------|
| <b>Recruiter Scheduling</b>  | Lets recruiters set the date, time, topic, and group size for a GD session.                |
| <b>Candidate Invitations</b> | Sends invitations to shortlisted candidates to join a specific GD session.                 |
| <b>Attendance</b>            | Tracks which invited candidates actually joined the session.                               |
| <b>Evaluation</b>            | Allows the recruiter or panel to score each participant on defined communication criteria. |
| <b>Shortlisting</b>          | Uses GD evaluation scores to determine which candidates progress to the next round.        |

## Module 15 — Interview Module

*Supports both AI-driven and traditional human-led interviews*

The Interview Module is the most personalized stage of the hiring pipeline, and HireSense AI supports two modes: a fully automated AI interview (powered by Modules 16–19) or a recruiter-scheduled human interview. This flexibility lets companies keep a human touch where they want it while still benefiting from automation at scale where they don't.

### Features (6)

|                                   |                                                                                                    |
|-----------------------------------|----------------------------------------------------------------------------------------------------|
| <b>AI Interview</b>               | An automated, chat or video-based interview conducted by the AI Question Generator and NLP engine. |
| <b>Human Interview Scheduling</b> | Lets recruiters book interview slots and share them with candidates for live interviews.           |
| <b>Interview Configuration</b>    | Defines interview length, question focus areas, and format for a given job.                        |
| <b>Recording</b>                  | Captures audio/video (or transcript) of the interview session for later review.                    |
| <b>Transcript</b>                 | Generates a text transcript of the interview conversation, especially for AI interviews.           |

|            |                                                                                        |
|------------|----------------------------------------------------------------------------------------|
| Evaluation | Produces or collects a structured evaluation of the candidate's interview performance. |
|------------|----------------------------------------------------------------------------------------|

## Module 16 — AI Question Generator

*Dynamically crafts relevant interview questions for each candidate*

Rather than relying on a static question bank, this module generates questions tailored to the specific candidate and role, drawing on resume content, job requirements, and core subject knowledge. This personalization makes AI interviews feel relevant and makes it harder for candidates to prepare generic, memorized answers.

### Features (6)

|                        |                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------|
| Resume-Based Questions | Generates questions directly referencing the candidate's listed projects, skills, and experience. |
| Role-Based Questions   | Generates questions aligned with the specific responsibilities of the job being applied for.      |
| Core Subject Questions | Draws from fundamental subject-matter question banks relevant to the candidate's field.           |
| Difficulty Levels      | Adjusts question difficulty based on the candidate's seniority or the role's requirements.        |
| Question Shuffling     | Randomizes question order and selection to reduce predictability across candidates.               |
| Adaptive Questions     | Adjusts subsequent questions based on the strength of previous answers.                           |

## Module 17 — NLP Follow-up Engine

*Makes AI interviews feel conversational by probing candidate answers*

This module gives the AI interview its conversational depth. Instead of moving through a fixed script, it analyzes what the candidate just said and decides whether a clarifying or probing follow-up is warranted — mimicking how a skilled human interviewer digs deeper into vague or interesting answers.

### Features (5)

|                       |                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------|
| Context Understanding | Maintains conversational context across the interview so follow-ups stay relevant.                  |
| Semantic Analysis     | Analyzes the meaning of a candidate's response, not just keywords, to gauge depth of understanding. |

**Dynamic Follow-Up Questions**

Generates a follow-up question in real time based on the candidate's previous answer.

**Response Analysis**

Evaluates the quality, relevance, and completeness of each response given.

**Maximum Follow-up Limit**

Caps how many follow-ups can be asked on a single topic to keep the interview on schedule.

## Module 18 — Fraud Detection System

*Safeguards the integrity of remote assessments*

Since assessments and interviews happen remotely and unsupervised, this module works in the background to detect behavior that suggests a candidate may be seeking unfair assistance. It doesn't automatically disqualify candidates but produces evidence and a fraud score that recruiters can factor into their decision.

### Features (5)

**Tab Switching Detection**

Detects when a candidate switches away from the assessment browser tab.

**Window Blur Detection**

Detects when the assessment window loses focus, a common sign of consulting another resource.

**Activity Monitoring**

Tracks general interaction patterns (keystrokes, mouse activity, idle time) during the assessment.

**Fraud Score**

Aggregates all suspicious signals into a single quantifiable score per candidate.

**Suspicious Behavior Logging**

Records a timestamped log of flagged events for recruiter review.

## Module 19 — OpenCV Monitoring

*Computer-vision-based presence and attention verification*

Working alongside the Fraud Detection System, OpenCV Monitoring uses the candidate's webcam feed to verify that the right person is present, alone, and attentive throughout the assessment or interview. This adds a visual layer of integrity checking that complements the browser-behavior signals from Module 18.

### Features (5)

**Face Detection**

Confirms a human face is present in the camera frame throughout the session.



|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Eye Tracking</b>            | Monitors gaze direction as a proxy for whether the candidate is looking at the screen.      |
| <b>Attention Monitoring</b>    | Flags sustained periods where the candidate appears distracted or absent from frame.        |
| <b>Multiple Face Detection</b> | Detects if more than one face appears in frame, which may indicate unauthorized assistance. |
| <b>Candidate Presence</b>      | Confirms continuous presence of the candidate for the full duration of the assessment.      |

## Module 20 — Evaluation Engine

*Combines every stage's results into one holistic assessment*

The Evaluation Engine is where all the individually collected scores — from resume, aptitude, coding, GD, and interview — come together. Using the weightage configuration set by the recruiter in Module 7, it computes a single, defensible final score and recommendation for every candidate, forming the analytical core of the platform's hiring decisions.

### Features (7)

|                             |                                                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------|
| <b>Resume Score</b>         | Pulls in the resume-matching score from Module 9 as one input to the final evaluation.               |
| <b>Aptitude Score</b>       | Incorporates the candidate's aptitude test result.                                                   |
| <b>Coding Score</b>         | Incorporates the candidate's coding assessment result.                                               |
| <b>GD Score</b>             | Incorporates the candidate's group discussion evaluation, if that round was enabled.                 |
| <b>Interview Score</b>      | Incorporates the candidate's interview evaluation, whether AI or human-conducted.                    |
| <b>Fraud Score</b>          | Factors in integrity signals from Modules 18–19 to flag or penalize suspicious behavior.             |
| <b>Final Recommendation</b> | Produces an overall recommendation (e.g., Hire, Hold, Reject) based on the weighted composite score. |

## Module 21 — Stage-wise Shortlisting

*Automatically advances or filters candidates after every round*

This module keeps the recruitment pipeline moving without manual intervention at every step. Immediately after a round concludes, it applies the recruiter's configured rules to automatically progress qualifying candidates and filter out those who don't meet the bar, keeping later rounds focused only on relevant candidates.

### Features (5)

|                                      |                                                                                             |
|--------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Cutoff-Based Filtering</b>        | Automatically eliminates candidates who fall below the configured score cutoff for a stage. |
| <b>Rank-Based Filtering</b>          | Advances only the top N ranked candidates from a round, regardless of absolute score.       |
| <b>Dynamic Candidate Progression</b> | Moves qualifying candidates into the next configured stage automatically.                   |
| <b>Automatic Rejection</b>           | Sends a rejection status (and typically a notification) to candidates who don't qualify.    |
| <b>Next Round Eligibility</b>        | Computes and displays which candidates are eligible for the upcoming round.                 |

## Module 22 — Report Generation

*Produces polished, shareable summaries of hiring outcomes*

This module turns raw evaluation data into human-readable reports for both students and recruiters. Candidate reports give students constructive feedback on their strengths and weaknesses, while recruiter reports summarize an entire hiring drive, both exportable as professional PDF documents.

### Features (8)

|                          |                                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------|
| <b>Candidate Report</b>  | A personalized report summarizing an individual candidate's performance across all rounds.  |
| <b>Recruiter Report</b>  | A summary report for recruiters covering all candidates and outcomes for a given job drive. |
| <b>Round-wise Scores</b> | Breaks down performance individually for each stage of the pipeline.                        |
| <b>Strengths</b>         | Highlights areas where the candidate performed particularly well.                           |
| <b>Weaknesses</b>        | Highlights areas where the candidate could improve, framed constructively.                  |
| <b>Fraud Summary</b>     | Summarizes any integrity flags raised during the candidate's assessments.                   |

**Hiring Recommendation**

States the system's final recommendation for the candidate.

**PDF Generation**

Exports the completed report as a downloadable, professionally formatted PDF document.

**Module 23 — Email & Notification System**

*Keeps every stakeholder informed automatically at every step*

Recruitment involves many time-sensitive touchpoints, and this module automates communication so no candidate or recruiter is left wondering about next steps. It spans both email delivery and in-app alerts, triggered automatically by events elsewhere in the system (registration, shortlisting, scheduling, results).

**Features (8)**

**OTP Emails**

Delivers one-time passwords for account verification and password resets.

**Welcome Emails**

Sends a welcome message upon successful registration.

**Job Notifications**

Alerts students about new job postings that match their profile or interests.

**Interview Invitations**

Notifies candidates when they are invited to an interview, with scheduling details.

**GD Invitations**

Notifies candidates when they are invited to a group discussion session.

**Result Emails**

Communicates round or final results to candidates automatically.

**Report Sharing**

Emails or shares a link to the generated candidate/recruiter report.

**In-App Notifications**

Surfaces real-time alerts within the platform's notification center.

**Module 24 — Analytics Dashboard**

*Turns platform activity into actionable insight*

This module transforms the raw operational data generated across the platform into visual, decision-ready insights for recruiters and administrators. It helps recruiters understand where candidates drop off in their pipeline and helps admins understand platform health and adoption trends.

**Features (6)**

**Hiring Funnel**

Visualizes candidate volume at each stage, from application through final selection.

**Candidate Statistics**

Summarizes candidate demographics, skill distribution, and performance trends.

|                    |                                                                                      |
|--------------------|--------------------------------------------------------------------------------------|
| Selection Rate     | Calculates the percentage of applicants who are ultimately selected.                 |
| Round Performance  | Shows aggregate pass/fail and score trends for each recruitment round.               |
| Recruitment Trends | Tracks hiring activity and outcomes over time to reveal seasonal or growth patterns. |
| Job Analytics      | Provides per-job metrics such as views, applications, and conversion rate.           |

## Module 25 — Workflow Engine (Core Innovation)

*Dynamically builds each job's recruitment pipeline in real time*

Positioned as the platform's core innovation, the Workflow Engine reads the configuration set in Module 7 and turns it into an actual, executable candidate pipeline — deciding which stages run, in what order, and under what conditions. This is what allows HireSense AI to support radically different hiring processes for different companies without hard-coding a single fixed flow.

### Features (5)

|                             |                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------|
| Configurable Workflow       | Builds the sequence of recruitment stages dynamically based on recruiter configuration.         |
| Conditional Round Execution | Runs or skips a round based on conditions such as candidate score or prior stage outcome.       |
| Round Skipping              | Allows certain candidates or entire jobs to bypass specific rounds when configured to do so.    |
| Pipeline Templates          | Reuses saved hiring templates to quickly instantiate a new job's workflow.                      |
| Workflow Visualization      | Displays the configured pipeline visually so recruiters can confirm the flow before publishing. |

## Module 26 — Security Module

*Protects platform data and infrastructure end to end*

Security is treated as a cross-cutting concern throughout HireSense AI, and this module consolidates the platform-wide protections that keep user data, credentials, and APIs safe from common threats. It works in tandem with Module 1's authentication layer but focuses more broadly on data protection and system hardening.

### Features (7)

|                     |                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------|
| Password Encryption | Stores passwords using strong one-way hashing (e.g., bcrypt) so plaintext is never persisted. |
|---------------------|-----------------------------------------------------------------------------------------------|

|                    |                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------|
| Secure File Upload | Validates and sanitizes uploaded files (resumes, logos) to prevent malicious payloads.     |
| Role Authorization | Enforces that every action is permitted for the requesting user's role before executing.   |
| Input Validation   | Sanitizes and validates all incoming data to prevent injection and malformed-data attacks. |
| API Protection     | Guards API endpoints against common attack vectors such as CSRF and unauthorized access.   |
| Rate Limiting      | Throttles repeated requests to prevent abuse such as brute-force login attempts.           |
| Audit Logs         | Maintains a tamper-evident record of security-relevant events for later review.            |

## Module 27 — File Management

*Organized, secure storage for every uploaded asset*

Across the platform, users upload many types of files — resumes, logos, reports, and profile photos. This module provides a consistent, centralized approach to storing, retrieving, and securing these assets, typically backed by cloud object storage for durability and scalability.

### Features (5)

|                  |                                                                                    |
|------------------|------------------------------------------------------------------------------------|
| Resume Storage   | Manages storage of student resume files with versioning and access control.        |
| Company Logos    | Stores recruiter-uploaded company branding assets for display across the platform. |
| Documents        | Handles miscellaneous supporting documents (e.g., certificates) uploaded by users. |
| Report Storage   | Persists generated PDF reports for later retrieval and sharing.                    |
| Profile Pictures | Manages profile photo uploads for students and recruiters.                         |

## Module 28 — Logging & Monitoring

*Provides visibility into system health and user behavior*

This module gives the engineering and admin teams the observability needed to run HireSense AI reliably in production. It captures everything from routine logins to unexpected errors, enabling faster debugging, better security auditing, and proactive performance management.

**Features (5)**

|                               |                                                                                            |
|-------------------------------|--------------------------------------------------------------------------------------------|
| <b>Login Logs</b>             | Records every login attempt, successful or failed, with timestamp and metadata.            |
| <b>Error Logs</b>             | Captures application errors and exceptions for debugging and reliability tracking.         |
| <b>Activity Logs</b>          | Logs significant user actions across the platform for traceability.                        |
| <b>Audit Logs</b>             | Maintains logs specifically for compliance and security auditing purposes.                 |
| <b>Performance Monitoring</b> | Tracks system metrics such as response time and resource usage to catch degradation early. |

## Module 29 — Future AI Services

*A roadmap of independent AI microservices to extend the platform*

This module outlines the platform's long-term AI roadmap, envisioning specialized microservices that can evolve and scale independently of the core application. Building these as separate services allows the team to iterate on AI models without disrupting the stability of the main platform.

**Features (5)**

|                          |                                                                                                  |
|--------------------------|--------------------------------------------------------------------------------------------------|
| <b>Resume AI</b>         | A dedicated service focused on advanced resume parsing, scoring, and matching.                   |
| <b>Interview AI</b>      | A dedicated service for conducting and evaluating AI-driven interviews at scale.                 |
| <b>Evaluation AI</b>     | A dedicated service for computing holistic, explainable candidate evaluations.                   |
| <b>Feedback AI</b>       | A dedicated service for generating personalized, constructive feedback for candidates.           |
| <b>Recommendation AI</b> | A dedicated service recommending suitable jobs to students or suitable candidates to recruiters. |

## Module 30 — Deployment & Scalability

*Production-grade infrastructure to support real-world hiring drives*

The final module covers the operational backbone that takes HireSense AI from a working prototype to a reliable, production-ready system. It focuses on containerization, environment management, and automated delivery pipelines so the platform can scale to handle large placement drives with many concurrent users.

**Features (6)**

|                                  |                                                                                              |
|----------------------------------|----------------------------------------------------------------------------------------------|
| <b>Docker Support</b>            | Packages the application and its dependencies into portable, consistent containers.          |
| <b>Environment Configuration</b> | Manages separate configurations for development, staging, and production environments.       |
| <b>Cloud Deployment</b>          | Supports deployment to cloud infrastructure for scalability and availability.                |
| <b>Database Backup</b>           | Automates regular backups of the database to prevent data loss.                              |
| <b>Monitoring</b>                | Provides infrastructure-level monitoring and alerting for uptime and performance.            |
| <b>CI/CD Ready</b>               | Supports continuous integration and continuous deployment pipelines for fast, safe releases. |

## 4. Suggested Technology Stack

The module specifications above are technology-agnostic by design. The stack below is a suggested starting point commonly used for platforms of this shape — feel free to substitute based on team familiarity and hosting constraints.

|                        |                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Frontend</b>        | React.js / Next.js, Tailwind CSS, Redux or React Query for state management                                   |
| <b>Backend</b>         | Node.js (Express/NestJS) or Django/FastAPI (Python), REST or GraphQL APIs                                     |
| <b>Database</b>        | PostgreSQL or MongoDB for core data; Redis for caching and session data                                       |
| <b>Authentication</b>  | JWT + Refresh Tokens, bcrypt for password hashing, OAuth for social login (optional)                          |
| <b>AI/NLP</b>          | OpenAI/LLM APIs or open-source LLMs for question generation and NLP follow-ups, spaCy/NLTK for resume parsing |
| <b>Computer Vision</b> | OpenCV + MediaPipe or a face-detection API for proctoring (Module 19)                                         |
| <b>Code Execution</b>  | Docker-based sandboxing, Judge0 or a custom container-per-submission runner                                   |
| <b>File Storage</b>    | AWS S3 / Cloudinary / Firebase Storage for resumes, logos, and reports                                        |
| <b>PDF Generation</b>  | reportlab / WeasyPrint (Python) or Puppeteer / pdf-lib (Node.js)                                              |
| <b>Notifications</b>   | SendGrid / AWS SES for email, Firebase Cloud Messaging for push/in-app alerts                                 |
| <b>DevOps</b>          | Docker, GitHub Actions for CI/CD, Nginx as reverse proxy, hosted on AWS/GCP/Azure                             |
| <b>Monitoring</b>      | Prometheus + Grafana or a hosted APM (e.g., Datadog, New Relic) for performance monitoring                    |



## 5. Suggested Build Roadmap

Given the scope of 30 modules, a phased build order is recommended so that each phase produces a working, demoable increment of the platform. Phases 1–3 establish the core marketplace; Phases 4–5 add the assessment and AI interview engine; Phases 6–7 close the loop with decisioning, reporting, and operational readiness; Phase 8 covers optional future AI enhancements.

### Phase 1 — Foundation

- Module 1: Authentication & Authorization
- Module 2: Landing Website
- Module 26: Security Module (baseline)

### Phase 2 — Core Profiles & Jobs

- Module 3: Student Management
- Module 4: Recruiter Management
- Module 5: Admin Panel
- Module 6: Job Portal
- Module 27: File Management

### Phase 3 — Workflow & Applications

- Module 7: Recruitment Workflow Configuration
- Module 10: Job Application Management
- Module 25: Workflow Engine
- Module 8: Resume Management

### Phase 4 — Assessments

- Module 11: Aptitude Test Module
- Module 12: Coding Assessment Module
- Module 13: Coding Sandbox
- Module 14: Group Discussion Module

### Phase 5 — AI Interview & Integrity

- Module 15: Interview Module
- Module 16: AI Question Generator
- Module 17: NLP Follow-up Engine
- Module 18: Fraud Detection System
- Module 19: OpenCV Monitoring

### Phase 6 — Evaluation & Insights

- Module 20: Evaluation Engine
- Module 21: Stage-wise Shortlisting
- Module 22: Report Generation
- Module 24: Analytics Dashboard

### Phase 7 — Communication & Ops

- Module 23: Email & Notification System
- Module 28: Logging & Monitoring
- Module 30: Deployment & Scalability

### Phase 8 — Future Enhancements

- Module 9: Resume Ranking AI (Future)
- Module 29: Future AI Services

## 6. Closing Notes

This report is intended as a working blueprint: each module's feature list can be converted directly into user stories, API endpoints, and database entities. The clearest dependency chain to keep in mind while planning sprints is Authentication (1) → Profiles (3, 4) → Job Portal (6) → Workflow Configuration (7) → Workflow Engine (25) → Assessment Modules (11–19) → Evaluation Engine (20) → Shortlisting (21) → Reports (22) & Analytics (24). Cross-cutting modules — Security (26), File Management (27), Logging (28), Notifications (23), and Deployment (30) — should be built incrementally alongside every phase rather than left until the end.

Module 9 (Resume Ranking AI) and Module 29 (Future AI Services) are explicitly marked as future/roadmap items in the source specification — they are documented here for completeness but can be safely deferred past the initial MVP without blocking core platform functionality.